A Derived Lorentzian-Type Signature from Observable Reactive Response in CTMT

The Negative Sign, Earned from Reactive Inversion,

with Dissipation Sign-Locked by the Second Law

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Abstract

We settle, within a precisely limited scope, the long-standing question of whether CTMT admits a *derived* indefinite (Lorentzian-type) signature rather than one inserted by hand. Working with the observable linear-response class $Z(\omega) = K - \omega^2 M + i\omega\Gamma$ (stiffness, inertia, damping; the RLC/coil/mechanical family used throughout CTMT) and its susceptibility $\chi = Z^{-1}$, we prove that the reactive (in-phase) response form $\mathrm{Herm}(\chi(\omega)) = \frac{1}{2}(\chi + \chi^\dagger)$ is an exact congruence of the real symmetric matrix $K - \omega^2 M$. By Sylvester's law of inertia its signature equals that of $K - \omega^2 M$: positive definite below the fundamental resonance, and *exactly one negative eigenvalue* on the frequency interval (ω_1, ω_2) between the first and second resonances—a Lorentzian $(1, n-1)$ signature. The negative direction is stiffness minus inertia going inertia-dominated: a difference of two genuinely positive physical terms, so the sign is earned, not inserted. The dissipative part is $\omega\, Z^{-1}\Gamma(Z^{-1})^\dagger \succeq 0$ by passivity (the second law), sign-locked and tied to the entropy-production result of the dynamics layer; the damping sets the width of the signature crossing. This is the honest realization of the throttle intuition: driven above its reactive bandwidth, the in-phase channel inverts. We state the boundary hard: this is *signature*, not curvature, and not general-relativistic dynamics; it is frequency-local, not a global metric. All results are exact and numerically verified for $n = 2, \ldots, 5$.

Contents

1 Purpose and the exact scope of the claim

The GR-facing question for CTMT has always reduced to one point: is there a logically defensible, non-inserted source for a negative metric sign? Earlier attempts placed the sign in the Fisher metric (impossible: it is a covariance, positive by construction) or in a rate–distortion Hessian with a hand-chosen opposite sign on the time term (an insertion). This note gives the sign a genuine home, on observable objects, and marks exactly how far it reaches.

Claimed and proved.
A derived indefinite *signature*: the reactive observable response form acquires one negative eigenvalue above the fundamental resonance, with the sign coming from a real physical competition (stiffness vs. inertia) and the dissipative part sign-locked by the second law.

Not claimed.
Curvature; a global spacetime metric; any general-relativistic dynamics (field equation, Bianchi identity, matter coupling). Those are separate structures this note does not touch (Section 6).

The result is therefore the scoped win: *the negative sign, earned*, not a theory of gravity.

2 The observable response class

CTMT's tested systems (coil/RLC, mechanical, diffusion-of-modes) share one linear-response form.

Definition 2.1 (Observable response). Let $K, M \in \mathbb{R}^{n\times n}$ be symmetric positive definite (stiffness, inertia) and $\Gamma \succeq 0$ symmetric (damping). The observable impedance and susceptibility at real drive frequency $\omega > 0$ are

$$Z(\omega) = K - \omega^2 M + i\omega\Gamma, \qquad \chi(\omega) = Z(\omega)^{-1}.$$

The *reactive* (in-phase) response form and the *dissipative* (absorptive) response form are

$$G_{\rm r}(\omega) := \mathrm{Herm}(\chi) = \tfrac{1}{2}(\chi + \chi^\dagger), \qquad G_{\rm d}(\omega) := -\tfrac{1}{2i}(\chi - \chi^\dagger).$$

The resonances are $\omega_1 \le \cdots \le \omega_n$ with $\omega_k^2 = \lambda_k(M^{-1}K)$.

These are physical, observable quantities: $G_{\rm r}$ is the response in phase with the drive (stored/reactive), $G_{\rm d}$ the response in quadrature (absorbed/dissipated).

3 The reactive signature theorem

Theorem 3.1 (Derived Lorentzian-type signature). *The reactive form is an exact congruence of the reactive impedance:*

$$\boxed{G_{\rm r}(\omega) = Z^{-1}\,(K - \omega^2 M)\,(Z^{-1})^\dagger.}$$

Consequently, by Sylvester's law of inertia, $\operatorname{sig} G_{\rm r}(\omega) = \operatorname{sig}(K - \omega^2 M)$*, and therefore:*

1. *for* $\omega < \omega_1$*,* $G_{\rm r} \succ 0$ *(Euclidean signature,* $(0, n)$ *negative count* 0*);*
2. *for* $\omega \in (\omega_1, \omega_2)$*,* $G_{\rm r}$ *has* exactly one *negative eigenvalue—a Lorentzian* $(1, n-1)$ *signature;*
3. *for* $\omega \in (\omega_k, \omega_{k+1})$*,* $G_{\rm r}$ *has exactly* k *negative eigenvalues.*

The negative directions are those in which the inertial term $\omega^2 M$ exceeds the stiffness K; the sign is thus the outcome of a competition between two positive physical forms, not an imposed convention.

Proof. Using $A^{-1} + B^{-1} = A^{-1}(B + A)B^{-1}$ with $A = Z$, $B = Z^\dagger$, and $Z + Z^\dagger = 2(K - \omega^2 M)$ (the $\pm i\omega\Gamma$ cancel),

$$\chi + \chi^\dagger = Z^{-1} + Z^{-\dagger} = Z^{-1}(Z^\dagger + Z)Z^{-\dagger} = 2\, Z^{-1}(K - \omega^2 M)(Z^{-1})^\dagger,$$

so $G_\mathrm{r} = Z^{-1}(K - \omega^2 M)(Z^{-1})^\dagger$. As Z^{-1} is invertible, this is a congruence; Sylvester's law of inertia gives $\operatorname{sig} G_\mathrm{r} = \operatorname{sig}(K - \omega^2 M)$. The eigenvalues of $K - \omega^2 M$ are positive for $\omega^2 < \omega_1^2$ and cross zero one at a time at each $\omega_k^2 = \lambda_k(M^{-1}K)$, giving the stated negative counts. Verified numerically for $n = 2, \ldots, 5$: $\operatorname{sig} G_\mathrm{r} = \operatorname{sig}(K - \omega^2 M)$ at every sampled frequency, with the single-negative window exactly (ω_1, ω_2). □

Remark 3.2 (Why this is the honest negative sign). The signature is not put in by choosing a metric convention; it is read off the inertia of $K - \omega^2 M$, i.e. from whether the driven response is stiffness- or inertia-dominated. Below resonance the system follows the drive in phase (all positive); above the first resonance one collective mode responds in antiphase—the reactive inversion familiar from every driven oscillator—and that mode is the single timelike direction. The minus sign is $K - \omega^2 M$: stored elastic response minus inertial response, a difference of two positive quantities.

4 Dissipation is sign-locked (the second law)

Proposition 4.1 (Passivity fixes the dissipative sign)**.** *For $\Gamma \succeq 0$ and $\omega > 0$,*

$$G_\mathrm{d}(\omega) = \omega\, Z^{-1}\Gamma\, (Z^{-1})^\dagger \;\succeq\; 0.$$

The dissipative response form is positive semidefinite at every frequency: the absorbed power is nonnegative (passivity / the second law). It is the damping Γ, not the reactive competition, and it sets the width over which the reactive signature crossing occurs.

Proof. $\chi - \chi^\dagger = Z^{-1}(Z^\dagger - Z)Z^{-\dagger} = Z^{-1}(-2i\omega\Gamma)Z^{-\dagger}$, so $G_\mathrm{d} = -\frac{1}{2i}(\chi - \chi^\dagger) = \omega Z^{-1}\Gamma(Z^{-1})^\dagger$, which is $\succeq 0$ since $\Gamma \succeq 0$. Verified numerically (min eigenvalue ≥ 0 across the sweep, all n). □

Remark 4.2 (Tie to the entropy-production lock). G_d is the observable, frequency-domain face of dissipation, and its fixed sign is the same second-law positivity that appears as the stationary entropy production $\sigma_\mathrm{EP} \geq 0$ of the dynamics layer. Reactive and dissipative responses thus play the two roles cleanly: the reactive part carries the (sign-changing) signature, the dissipative part is sign-locked and physical, and the damping scale is the throttle width. The negative sign lives entirely in the reactive competition; the arrow of time lives entirely in the dissipative positivity.

5 The throttle: the intuition, made exact

Corollary 5.1 (Throttle induces the sign)**.** *Increasing the drive frequency through the reactive bandwidth (equivalently, narrowing the channel until the drive outruns its reactive response) carries G_r from positive definite to indefinite: at $\omega = \omega_1$ the first eigenvalue crosses zero and becomes negative. The onset is exactly the point where inertia overtakes stiffness on the softest mode.*

This is the rigorous content of the original intuition—"information forced through a throttled place produces the sign." The precise mechanism is reactive inversion: past the reactive bandwidth the in-phase response changes sign. The throttle produces *signature*. It does not, by itself, produce curvature (Section 6).

6 Scope: signature, not curvature, not dynamics

The boundary must be stated as firmly as the result.

Non-claim 6.1 (Signature is not curvature). Theorem 3.1 yields an indefinite *signature* of a response form at a fixed probe frequency. Curvature is a property of how such a form varies over a base manifold, through a connection with nonzero holonomy. That is the open bundle problem (Dynamics I, Rungs 4–5) and is *not* established here. A negative eigenvalue is a necessary ingredient of a Lorentzian geometry, not a Lorentzian geometry.

Non-claim 6.2 (No general-relativistic dynamics). Nothing here provides a field equation, Bianchi identity, stress–energy coupling, or diffeomorphism dynamics. The signature does not obey, and is not claimed to obey, the Einstein equations. This note does not attempt GR and does not bear on whether CTMT could address problems in GR.

Non-claim 6.3 (Frequency-local, not global). The signature depends on ω: it is $(0, n)$ below ω_1, Lorentzian $(1, n-1)$ on (ω_1, ω_2), and higher-index above. There is no single global signature; the Lorentzian regime is the specific window (ω_1, ω_2).

Remark 6.4 (What is nonetheless genuinely gained). Within these limits the gain is real and was previously missing: a negative metric sign that is *derived* (Sylvester congruence to $K - \omega^2 M$), *physical* (reactive inversion of an observable response), and *clean* (exactly Lorentzian on a definite frequency window), with the companion dissipative sign fixed by the second law. The sign is no longer an assumption of CTMT; it is a theorem about observable response.

7 Conclusion

The question was whether CTMT has any defensible, non-inserted source for the negative sign. It does: the reactive part of the observable response form is congruent to $K - \omega^2 M$, so its inertia flips one eigenvalue negative between the first and second resonances, producing a Lorentzian $(1, n-1)$ signature; the dissipative part is sign-locked by passivity and carries the second-law positivity of the entropy-production layer; and the transition is driven by the throttle exactly as intuited. The minus sign is stiffness minus inertia—two positive physical terms in competition—not a chosen convention.

This is the scoped win, and its boundary is the honest place to set the GR path down. What is secured is *signature*: an emergent Lorentzian sign on observable response, earned rather than assumed. What remains open, and is not claimed, is curvature (the moving-form bundle problem) and, beyond it, any gravitational dynamics. CTMT now has a physical negative sign; it does not have, and this note does not seek, general relativity. That is a defensible flag to plant—the sign is real, the ceiling is stated, and neither is overclaimed.

References

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